



Designing Authentic Industry-Engaged Assessment for Professional Competence in Business Intelligence

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INDUSTRIAL ENGINEERING

Academia

Aims to prepare students for professional practice, but offers limited guidance on how to operationalize this in course design and assessment.

Industry

Expects credible, reproducible results supported by clear reasoning and usable for decision-making.

Contrasting evaluation perspectives (cause for tension)

-  methods and iteration (errors acceptable if reasoning is sound)
-  credibility in context (early errors can undermine trust)

Core challenge

How do we assess **professional competence** in team-based, open-ended projects?

Particularly:

- ▶ collaboration
- ▶ reasoning and justification
- ▶ reproducibility

Standard assessment:

- ▶ often focuses on outputs
- ▶ weak visibility into process and contribution

Overview

- ▶ 6 ECTS course (advanced undergraduate / early graduate)
- ▶ 10–15 students, two teams
- ▶ Semester-long project with **industry partner**
- ▶ Five modules + Capstone

What is BI?

Data-driven decision support: combining data analysis, systems, and communication to inform organizational decisions.

Goal

Strengthen industrial relevance and professional competence

- ▶ CDIO-aligned redesign
- ▶ Assessment as **developmental mechanism**



Key idea

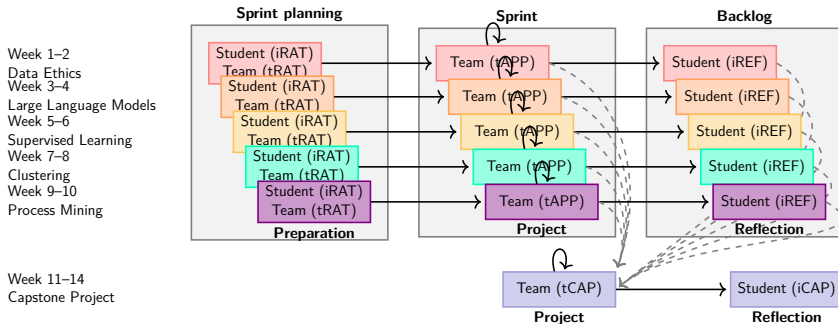
+ feedback
iteration
⇒ competence

LOUIS framework

Learning Outcomes in University for Impact on Society
Focus on a small set of competences (max 3)

Civic engagement	Creative thinking	Critical thinking	Ethical reasoning
Global learning	Information literacy	Inquiry and analysis	Integrative learning
Intercultural knowledge & competence	Foundations for life-long learning	Oral communication	Problem solving
Quantitative literacy	Reading	Teamwork	Written communication

Pedagogical structure



 Team-Based Learning (TBL):

-  iRAT (individual readiness)
-  tRAT (team readiness)
-  tAPP (team application)
-  iREF (individual reflection)

 Flipped classroom

 Iterative modules leading into capstone

Core mechanism: assessment as development

☰ One rubric, many iterations

Same assessment rubric reused across all modules and the capstone

Standards fixed; performance expectations scaled during learning

☑ Assessment criteria

- ▶ analytical soundness
- ▶ transparency
- ▶ reproducibility
- ▶ stakeholder relevance

▶▶ Developmental shift

Feedback is **carried forward**
— *not* reset between tasks

Pull Requests as assessment architecture

Changes are proposed, reviewed, discussed, and integrated – or explicitly deferred

Structured accumulation

- ▶ same rubric reused
- ▶ prior feedback revisited
- ▶ unresolved issues explicitly carried forward

Enabled by tooling

-  version control
-  persistent work traces

Sustained industrial context

A single dataset and domain used across the entire semester

Industry partner role

- ▶ problem framing
- ▶ milestone discussions
- ▶ final evaluation

Key difference

Academic vs. industry evaluation criteria are **not the same**

Industry threshold

Credibility is fragile: early analytical flaws can undermine trust

Explicit evaluation criteria

- ▶ reproducibility
- ▶ transparency
- ▶ justification of analytical choices

Student insight

Sound reasoning is necessary
– but not sufficient

Team-level assessment

- ▶ analytical quality
- ▶ coherence
- ▶ relevance

Individual-level assessment

- ▶ visible artifacts
- ▶ documented contributions
- ▶ structured reflection

Design goal

Accountability without fragmenting teamwork

Designed into the course architecture—not added after the fact

Multiple feedback sources

Students learn to navigate intentionally different evaluation criteria

Instructor

- ▶ method
- ▶ implementation

Industry partner

- ▶ domain relevance
- ▶ decision context

Student task

Adapt analysis and communication to audience

Students & graduates

- ▶ high engagement without attendance requirements
- ▶ transfer of reproducible workflows
- ▶ stronger documentation practices

Academic and industry perspectives

-  more ambitious, student-driven projects
-  improved coherence and credibility

Inherent design trade-offs

These tensions are not eliminated – they are deliberately designed for

- ▶ development vs. reliability
- ▶ teamwork vs. individual accountability
- ▶ authentic tools vs. cognitive load

Transferable principles

For CDIO-aligned, industry-engaged courses

- ▶ sustained industry context
- ▶ assessment as development
- ▶ visible individual contribution
- ▶ forward-oriented reflection
- ▶ explicit management of expectation gaps
- ▶ scaffolded professional tooling

Scope and constraints

This is a design-oriented case, not a controlled experiment

Current limitations

- ▶ depends on student readiness
- ▶ scalability considerations
- ▶ domain-specific transferability

Design refinements

- ▶ earlier preparation for professional tooling
- ▶ improved stakeholder interaction
- ▶ structured team-building to accelerate trust in large groups
- ▶ scaling across courses and partners

What this case shows

Assessment and feedback can actively structure professional competence development

- ▶ industry engagement surfaces critical evaluation criteria
- ▶ credibility and trust require explicit design attention
- ▶ cumulative, iterative feedback supports deeper learning

Core contribution

Assessment architecture—*not specific tools*—enables professional learning

Questions?



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